

The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 4 MODULE, AUTUMN SEMESTER 2014-2015

G54VIS COMPUTER VISION

Time allowed TWO Hours

Candidates must NOT start writing their answers until told to do so

Answer THREE out of FOUR questions

No calculators are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

1. Answer all of the following sub-questions pertaining to Edge Detection, Segmentation, and Gaussian Mixture Models (20 marks):

- (a) Edges in an image often represent sharp changes in image properties such as the intensity of a gray-scale image. Describe what an image gradient is and explain in what manner it is related to the edge and how it can be used to find edges in a gray-scale image.

[6 marks]

- (b) The Canny edge detector has a key step of keeping edges connected after thresholding, which is known as thresholding with Hysteresis. Explain how this concept works. You may also draw a graph to illustrate it.

[5 marks]

- (c) Consider the following 4x4 map of intensity gradients. Apply the watershed algorithm to this map of gradients, clearly labeling each step of the process. As a final result you should show a segmentation with one or more labeled regions. (Note that watersheds are not considered a labeled region.)

[11 marks]

1	3	1	3
1	3	1	1
2	2	2	1
1	1	1	1

- (d) Consider modelling the colour distribution of an image with a Gaussian Mixture Model (GMM), given as

$$p(\mathbf{x}) = \sum_{i=1}^n \pi_i p(\mathbf{x} | \theta_i)$$

where $\mathbf{x}$ is the colour information, n the number of different Gaussians, $p(\mathbf{x} | \theta_i)$ the individual Gaussian function, θ_i the Gaussian parameter, and π_i the weight for i -th Gaussian. Explain the meaning of θ_i and π_i , and how the above equation models the colour distribution of a given image. You may draw a graph for illustration.

[6 marks]

- (e) Both K-means and GMM employ iterative schemes for image segmentation. Briefly describe the general procedure of the iterative algorithm and explain the difference between the concepts of K-means and GMM for image segmentation problems.

[5 marks]

2. Answer all of the following sub-questions pertaining to Motion estimation and Optical Flow (20 marks):

- (a) List three possible simple techniques (not including optical flow) for motion estimation between two successive frames, and explain how each of them works.

[4 marks]

- (b) Compare and contrast the concepts of "motion field" and "optical flow".

[4 marks]

- (c) Describe the two main assumptions made in the Horn-Schunk algorithm, and explain how they are combined when estimating the optical flow between two successive frames in a video sequence.

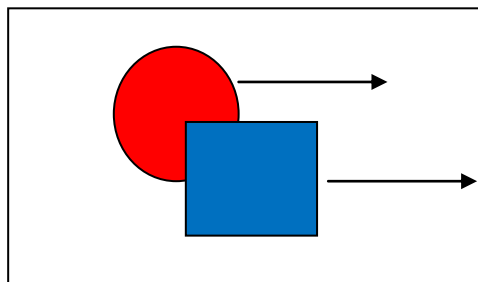
[7 marks]

- (d) Consider the situation where you were sitting in a driving car and had been taking a video of the street at one side of the car. There are trees alongside the street and the houses located further back. Address at least one problem if you were to use Horn-Schunk algorithm to estimate the optical flow between the video frames. Explain how the problem(s) may be dealt with efficiently.

[11 marks]

- (e) Consider the following two moving objects, a circle and a square, with a static background. They are moving towards the right at the same speed. Locate where the motion flow discontinuity would happen. Justify your answer.

[7 marks]



3. Answer all of the following sub-questions pertaining mainly to Tracking, Mean Shift, and Particle Filters (20 marks):

- (a) A visual tracking system consists of three main components, an appearance model, a motion model, and a search engine/framework. Explain each of the components and their functionalities.

[4 marks]

- (b) Consider the following likelihood map. Explain how a tracker may move from the circled pixel to the mode (the maximum likelihood value), based on the Mean Shift algorithm using a 3×3 window. Specify the formula that you will be using for computing the mean shift vector. Discuss any issues you may encounter and their possible solutions.

[11 marks]

1	1	1	1	1	1	1
1	2	2	2	2	1	1
1	1	2	3	3	3	1
1	1	2	3	4	3	2
1	1	1	2	3	3	2
1	1	1	1	2	2	2
1	1	1	1	1	1	1

- (c) List at least three problems that a tracker may face when tracking objects/subjects in a video. Explain how they may complicate tracking problems.

[4 marks]

- (d) What is the main difference between the assumption of the Mean Shift algorithm and that of the particle filter for motion tracking? Give one exemplar scenario for each of the methods.

[7 marks]

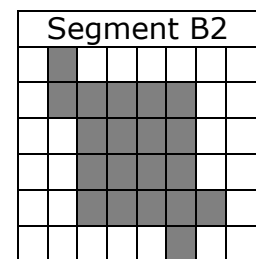
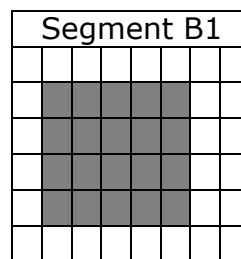
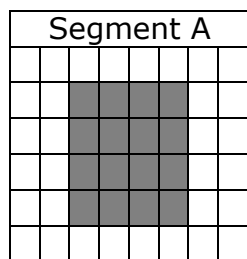
- (e) A particle filter employs a non-parametric method for approximating the probability density of the current state of a target, based on the measurements of the target at all previous times. Explain the meaning of a particle and how it is used to predict the state of the target at the next frame.

[7 marks]

4. Answer all of the following sub-questions pertaining primarily to Evaluation of Computer Vision algorithms, scene analysis and Hidden Markov Models (20 marks):

- (a) Consider the three segments below (in dark grey). Consider the "Earth mover" distance for computing the distance between segments B1 and A, and segments B2 and A. Consider another distance metric of your choice and compute the distances between the same pairs of segments (i.e. distance(B1,A) and distance(B2,A)). Discuss the pros and cons of the two different distance measures.

[11 marks]



- (b) Describe three ways to select test data for evaluating any Computer Vision algorithms. Explain the pros and cons for each of them.

[4 marks]

- (c) A Hidden Markov Model (HMM) has two sets of variables (i.e., hidden states, and observations) and three sets of functions (i.e., start state functions, state transition functions, and observation functions). Based on this, describe three typical problems that a HMM may solve.

[7 marks]

- (d) Consider a "one-way highway tunnel scene", whereby cars typically drive in from one end of the tunnel, and after a certain amount of time drive out from the other end. How would you propose to use a Hidden Markov Model (HMM) for modeling this scene in general, and for detecting any abnormal behaviour in particular?

[11 marks]